

Internet of Things

Data propagation algorithms III

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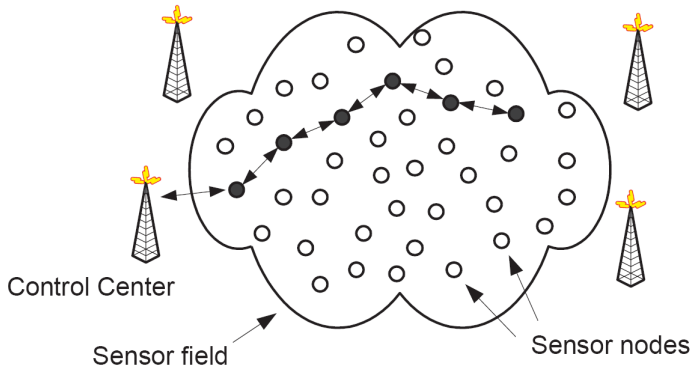
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Previously on WSAN ...



The problem

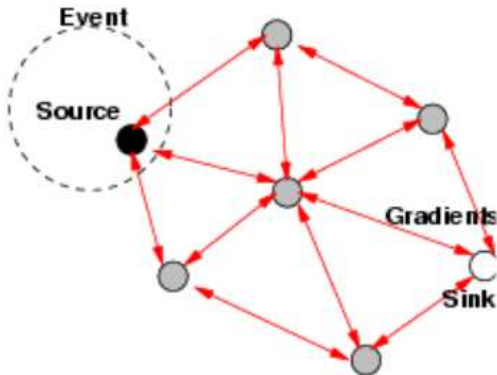
“How can sensor p , via cooperation with the rest of the sensors in the network, propagate information about event $\mathcal{E}$ to the control center(s)?”





Directed Diffusion

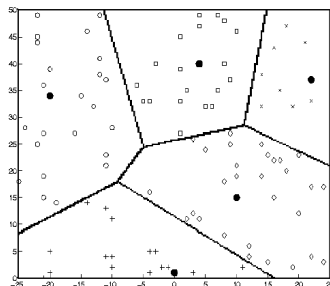
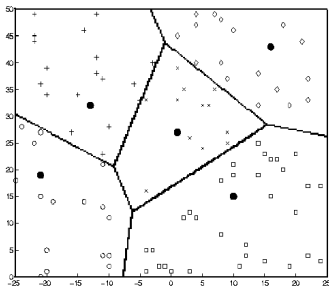
- Data-centric protocol
- Can handle low dynamics





LEACH

- Scalable, energy efficient protocol employing randomization





Next, on Internet of Things...

A. Link quality based methods

- ETX metric*
- Minimum Outage Rate (MOR)*

B. Link distance based methods

- Routing with relay diversity (ExOR)*
- Greedy, local neighbour selection (LTP)

C. Multipath routing

- Braided paths (GRAd, GRAB)
- Probabilistic forwarding (PFR)

* Protocols we will focus on



A. Link Quality Methods

- **Main goal:** Providing robustness by selecting routes that *minimize end-to-end re-transmissions or failure probabilities*.
- **Motivation:** If wireless links were (ideally) error-free, then a shortest hop-count path could be chosen. Such paths however require long distance links, which in realistic settings are likely to be error-prone. Thus, the link quality should be taken into account (via simple ACK signals for each successfully delivered packet).



A1. The ETX metric

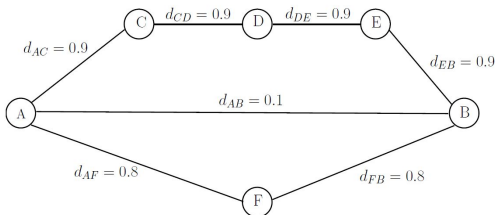
- Let d_f be the **packet reception rate** (probability of successful delivery) over a link in the forward direction, and d_r the **probability that the corresponding ACK is received** in the reverse direction.
- Assuming each packet transmission can be abstracted as a Bernoulli trial, **the expected number of re-transmissions** required for successful delivery of a packet on this link is:

$$ETX = \frac{1}{d_f d_r}$$

- This metric for a single link can then be incorporated into any routing protocol, so that chosen end-to-end paths **minimize the sum of ETX over all links on the path**, i.e. the total expected number of transmissions on the path.



ETX routing - example



- This example shows 3 routes from A to B, with the forward probabilities on each link (for simplicity, all reverse probabilities are taken $d_r = 1$).
- The direct A-B transmission would require an expected number of 10 re-transmissions.
- The path via C,D,E would require $4(1/0.9) \simeq 4.44$ re-transmissions (1.1 per link) on average.
- The A-F-B path would require $2(1/0.8) \simeq 2.5$ re-transmissions (1.25 per link) on average.

$\Rightarrow$ the ETX-minimizing path is A-F-B.

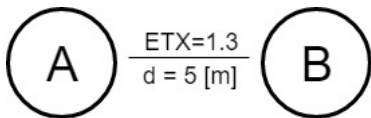


ETX routing - discussion

- As demonstrated in the example, ETX *neither favours* long paths with many short-distance, high-quality links, *nor very short paths* with a few long-distance, low-quality links. It actually takes an "*in-between*" approach.
- **ETX advantages:** It minimizes the number of transmissions required, as well as it improves energy efficiency. Also, it directly addresses the potential link asymmetries, by taking into account packet delivery probabilities in both directions.
- **ETX limitation:** *It assumes knowledge* of the d_r , d_f packet reception probabilities. In quite static networks these values may be periodically obtained via link monitoring, however in dynamic networks the obtained values may become obsolete very soon and the link monitoring overhead may be prohibitively high.



ETX - Example 2



Assuming that:

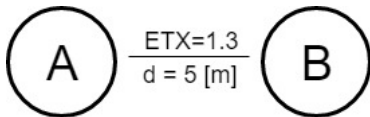
- The energy required to operate the radio module for each bit (both for receiving and transmitting) is $E_c = 50$ [nJ/bit];
- The energy required to successfully transmit a message over distance d is $E_{tx}(d) = kd^2$ [nJ/bit], where $k = 1$;

Calculate:

The expected overall network energy consumption for routing one packet of length $b = 100$ [bits] from A to B.



ETX - Example 2



Assumptions:

- $E_c = 50[nJ/bit]$;
- $E_{tx}(d) = kd^2[nJ/bit]$, where $k = 1$;
- $b = 100[bits]$.

Answer:

$$\begin{aligned} E_{A-B} &= 1.3 \cdot (E_c \cdot b + E_{tx}(5) \cdot b + E_c \cdot b) \\ &= 1.3 \cdot (50[nJ/bit] \cdot 100[bit] + \\ &\quad + 5^2[nJ/bit] \cdot 100[bit] + 50[nJ/bit] \cdot 100[bit]) \\ &= 16,250nJ = 16.25\mu J \end{aligned} \tag{1}$$



A2. The Minimum Outage Route (MOR) Metric (I)

- Under *high dynamics* (mobile nodes, mobile objects) the link quality fluctuates very rapidly and *ETX is not feasible*. In such cases, *analytic models* of link quality can be used. They explicitly model the wireless channel as having multi-path fading with Rayleigh statistics (fluctuations over time).
- Let d be the distance between transmitter and receiver, h the path-loss exponent, SNR the signal-to-noise ratio without fading, f the fading state of the channel. Then, the *instantaneous* capacity of the channel is:

$$C = \log \left(1 + \frac{|f|^2}{d^h} \text{SNR} \right)$$



A2. The Minimum Outage Route (MOR) Metric (II)

- The *outage probability* P_{out} is defined as the probability that the instantaneous channel capacity falls below the transmission rate R . It is shown that:

$$P_{out} = 1 - \exp\left(\frac{-d^h}{\mu SNR^*}\right)$$

where $\mu = E[|f|^2]$ is the mean of the Rayleigh fading and SNR^* a normalized SNR.



A2. The Minimum Outage Route (MOR) Metric (III)

- Then, end-to-end reliability of a route is defined as the probability that *none of the intermediate links suffers outage*, and the most reliable route between two nodes is one *that minimizes the following path metric*:

$$\sum_i d_i^h$$

where d_i is the distance of the i^{th} hop in the path. The metric d^h for each link is called the minimum outage route (MOR) metric.

- The MOR approach does not require collection of link quality metrics like ETX, neither ACK messages are used. However, *its practicality is limited by the basic abstraction that the channel fading follows a certain distribution* (the Rayleigh distribution).

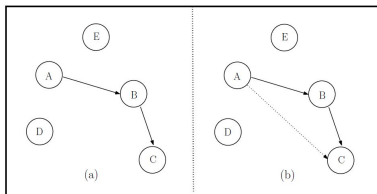


B. Link Distance Methods

- In general, nodes closer *to the data destination* are chosen/favoured.
- The relay diversity is provided by the fact that wireless transmissions are *broadcast to multiple* nodes.



B1. Routing With Relay Diversity

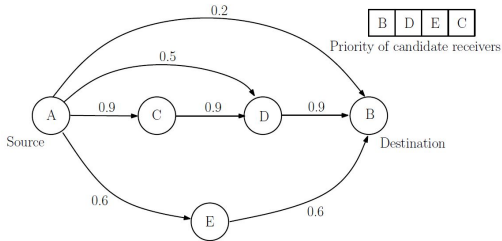


- In traditional routing, the reliability of the AC path depends on whether both AB and BC transmissions have been successful.
- However, if *C is also allowed to accept packets directly from A*, then the reliability can be further increased without much additional energy cost.
- Allowing such 2-hop packet reception, in the high SNR regime, *the end-to-end outage probability decays as $(\text{SNR})^{-2}$* . When nodes within L hops can communicate with each other with high SNR, this probability can become as small as $(\text{SNR})^{-L}$.
- A weakness of this method is that it requires *a larger number of receivers to be actively overhearing* each message, which may incur a *radio energy penalty*.



Extremely Opportunistic Routing (ExOR) (I)

- **Main idea:** the identity of the node which will eventually forward a packet, *is not predetermined before the packet is transmitted*. Instead, the method tends to ensure that *the node closest to the destination receiving a given packet will forward the packet further*.



- The protocol has *three stages*:
 1. **Priority ordering:** At each step, the transmitter includes in the packet a schedule with the priority order of candidate receivers that should forward the packet.



Extremely Opportunistic Routing (ExOR) (II)

2. **Transmission acknowledgements:** A MAC scheme is used so that each candidate receiver sends the ID of the highest-priority successful recipient known to it. All nodes listen to all ACKs, so they *distributively determine which node, among those who received the packet successfully, has the highest priority.*
 3. **Forwarding decision:** After listening to all ACKs, the nodes that have not heard of any IDs with priorities greater than their own will transmit.
- Features of the ExOR protocol
 - *Nodes further away from the current node (yet closer to the destination) are less likely to successfully receive the packet, but, whenever they receive it, they are favoured to act as forwarders!*
 - *This tends to make good progress towards the data destination, without many transmissions and delays.*
 - As with relay diversity, ExOR **requires a larger number of receivers** to be active. Also, the priority evaluation necessitates some inter-node packet delivery ratios to be tracked and maintained.



B2. Greedy, Local Neighbour Selection (LTP)

Network nodes are called “ *particles*”.

- a) Each particle has *two communication modes*: a *broadcast* (radio) *beacon mode* and a *directed point-to-point* transmission mode (unicast mode).
- b) Each particle may alternate between a **sleeping** and an **awake** mode. During sleeping periods particles cease any communication.
- c) Particles *do not move*.
- d) The particles are spread in a two-dimensional area (plane).



LTP model continued

- e) A *receiving wall* $\mathcal{W}$ is a *line* in the plane. The wall represents the control center.
- f) Each particle is *aware of the direction* toward $\mathcal{W}$.
- g) *No geolocation* abilities assumed

Definitions: Let d (in numbers of particles / m^2) be the **density** of the cloud.

Let $\mathcal{R}$ be the maximum (beacon/laser) **transmission range** of each particle.

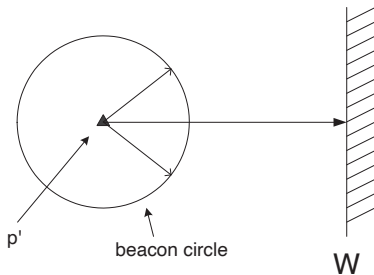


The Local Target Protocol (LTP)

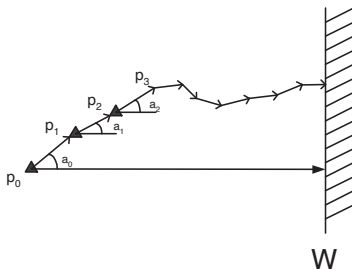
Let $d(p_i, p_j)$ the (vertical) distance of p_i, p_j and $d(p_i, \mathcal{W})$ the (vertical) distance of p_i from $\mathcal{W}$. Let **info**($\mathcal{E}$) the info to be propagated. Each p' receiving **info**($\mathcal{E}$) does the following:

- **Search Phase:** It uses a low energy broadcast of a beacon (angle a above and below the vertical line) to *discover* a particle closer to $\mathcal{W}$ (i.e. a p'' where $d(p'', \mathcal{W}) < d(p', \mathcal{W})$).
- **Direct Transmission Phase:** If found, p' *sends* **info**($\mathcal{E}$) to p'' via a unicast transmission.
- **Backtrack Phase:** If repetitions of the search phase *fail* to discover a particle nearer to $\mathcal{W}$, then p' *sends* **info**($\mathcal{E}$) *to the particle it received the information from*.

Example of the Search Phase



Example of a Transmission





Efficiency

Definitions: Let h_{opt} the (*optimal*) number of “hops” (**vertical to $\mathcal{W}$** transmissions) needed to reach $\mathcal{W}$, if particles always exist in pair-wise distances $\mathcal{R}$ towards $\mathcal{W}$.

Let h the *actual* number of hops (transmissions) taken to reach $\mathcal{W}$. The “**hops**” **efficiency** of the data propagation protocol is the ratio

$$C_h = \frac{h}{h_{opt}}$$

where $h_{opt} = \left\lceil \frac{d(p, \mathcal{W})}{\mathcal{R}} \right\rceil$



Why studying h , C_h ?

When a particle p “looks around” for a particle as close to $\mathcal{W}$ as possible to pass information, it may not get any particle in the perfect direction (on the line vertical to $\mathcal{W}$ passing from p), mainly because:

- a) There might never have been any particles in that direction.
- b) Particles of sufficient remaining battery power may not be available.
- c) Particles available may temporarily “sleep” to save energy.



Summary evaluation of LTP

- local, simple, greedy protocol
- no global structure (set of paths) maintained
- good for dense networks
- performance drops in sparse / faulty networks



C1. Multipath Routing

- multiple routes are used, towards increasing robustness
- the routes can be *disjoint* or *partially disjoint*
- "Braided multipath routing":
 - There is a primary path that is used for routing.
 - Several alternative paths are maintained for use in case of a failure in the primary path.
 - "*Braided*" path: node disjointness between alternative paths is not a strict requirement. Rather, for each node on the primary path, the method requires the existence of an alternative path from the source to the sink *that does not contain that node*, but which may otherwise overlap with the other nodes on the main path.
 - Compared to disjoint paths, the braided path is more suitable for *isolated failures*, while disjoint paths are more appropriate for pattern (*geographically correlated*) failures.



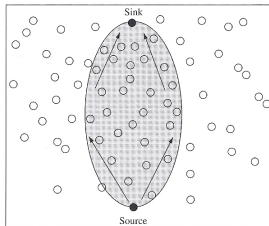
Gradient Cost Routing (GRAd)

- All nodes in the network maintain an estimated **cost to the sink** (such as the number of hops needed to reach it).
- When a packet is transmitted, it includes a field with the **cost "paid"** already in the current data propagation. Also, a **remaining value** field is maintained, acting as a TTL (time-to-live) field for that packet.
- Any receiver that receives this packet forwards the packet **iff its own cost to the sink is smaller than the remaining value of the packet**. Before forwarding, the "cost paid" field is increased by one and the remaining value field is decreased by one.
- GRAd actually **allows *multiple* nodes to forward the same message**, so it essentially performs *a limited directed flooding* towards the sink and provides significant robustness but at the cost of a larger overhead.



Gradient Broadcast Routing (GRAB)

- It enhances GRAd by **incorporating a tunable energy-robustness trade-off** through the use of credits.
- Similarly to GRAd, GRAB maintains a cost field at each node. Additionally, the packets travel from a source to the sink with a **credit value** that is decreased at each step depending on the hop cost.
- **Credit-sharing mechanism:** *Earlier hops receive a larger share of the total credit in a packet, while the later nodes receive a smaller share. Intermediate nodes with greater credit can spend a larger budget sending the packet to a larger set of forwarding neighbours. This way, paths *spread out initially* while eventually the *diverse paths converge to the sink* efficiently.*





C2. The Probabilistic Forwarding Protocol (PFR)

We assume that each node (particle) has the following abilities:

- i) It can estimate **the direction of a received transmission** (e.g. via the technology of direction-sensing antennae).
- ii) It can estimate **the distance** from a nearby particle that did the transmission (e.g. via estimation of the attenuation of the received signal).
- iii) It knows the **direction towards the sink S**. This can be implemented during a set-up phase, where the sink broadcasts information about itself to all particles.
- iv) All particles have a common **coordinates system**.

Note that *GPS information is not needed* for this protocol. Also, there is no need to know the global structure of the network.



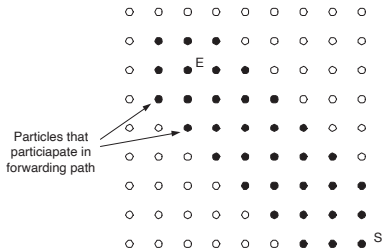
Propagation Protocol Properties

- **Correctness.** Protocol Π must guarantee that *data arrives to the sink* S , given that the network is operational.
- **Robustness.** Protocol Π must guarantee that data arrives at enough points in a small interval around S , in cases where part of the network has become inoperative.
- **Efficiency.** Protocol Π should have a *small ratio* of the number k of *activated* particles over the total number N of particles $r = \frac{k}{N}$. Thus r is an energy efficiency metric of Π .



The basic idea of PFR

PFR *probabilistically favours* transmissions toward the sink within a *thin zone* of particles around the line connecting the particle sensing the event $\mathcal{E}$ and the sink.



Thin Zone of particles

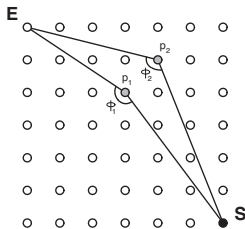


The Forwarding Probability

Data is propagated with a suitably chosen *probability* p , while it is not propagated with probability $1 - p$.

To *favour near-optimal transmissions* we choose

$$P_{fwd} = \frac{\phi}{\pi}$$





The two phases of PFR

Phase 1: The “Front” Creation Phase. Initially a *sufficiently large “front” of particles* is built (by using a limited, in terms of rounds, flooding), to guarantee the survivability of the data propagation process. Each particle having received the data, *deterministically broadcasts* them toward the sink.

Phase 2: The Probabilistic Forwarding Phase.

Each particle p receiving $\text{info}(\epsilon)$, broadcasts it to all its neighbours with probability $\mathbb{P}_{fwd}$ (or it does not propagate any data with probability $1 - \mathbb{P}_{fwd}$) defined as follows:

$$\mathbb{P}_{fwd} = \begin{cases} 1 & \text{if } \phi \geq \phi_{threshold} = 134^\circ \\ \frac{\phi}{\pi} & \text{otherwise} \end{cases}$$



The Correctness of PFR

Lemma

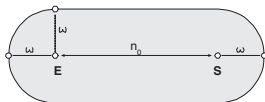
PFR *always succeeds* in sending the information from E to S when the whole network is operational.

In the proof, geometry is used (i.e. we *cover* the network area by unit squares and show that there are always particles “*close enough*” to the *optimal line*, i.e. with $\phi > 134^\circ$, that deterministically broadcast).



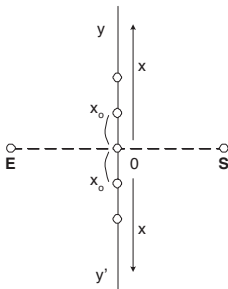
The Energy Efficiency of PFR

- Consider particles that are active but *as far as possible* from ES



The particles inside the L_Q Area

- w is *approximated* with the following random walk





The Energy Efficiency of PFR

By using stochastic dominance by a continuous time “discouraged arrivals” birth-death process, the following is proved:

Theorem

The energy efficiency of the PFR protocol is $\Theta\left(\left(\frac{n_0}{n}\right)^2\right)$ where $n_0 = |ES|$ and the total number of particles in the network is $N = n^2$. For $n_0 = o(n)$, this is $o(1)$.



The Robustness of PFR

Particles *very near the ES line* are considered.

Consider the case when *some* of these particles (at angles $> 134^\circ$) are *not operating*.

The probability that none of them transmits is *very small*.

Authors show:

Lemma

PFR manages to propagate the crucial data across lines parallel to *ES*, and of constant distance, with *fixed* nonzero probability.



Experimental Comparison of LTP and PFR

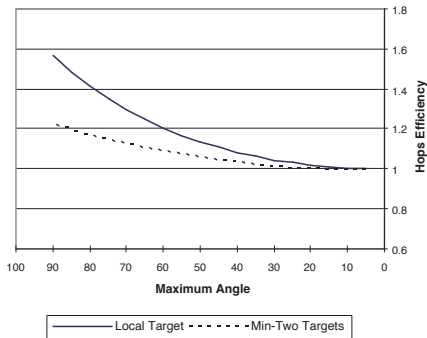
- C++
- 2D geometry data types of LEDA
- a variety of sensor fields in a $100\text{m} \times 100\text{m}$ square area:
 - we drop randomly $n \in [100, 3000]$ particles (i.e. $0.01 \leq d \leq 0.3$)
 - fixed radio range $R = 5\text{m}$
 - search angle $a = 90^\circ$
- each experiment was repeated for more than 5000 times to get good average results



LTP – The impact of angle α

- $\alpha \rightarrow 0 \Rightarrow C_h \rightarrow 1$
- C_h initially decreases very fast, while having a limiting behavior for $\alpha \leq 40$

Ideal Hops Efficiency for angles $\alpha \in [5, 90]$

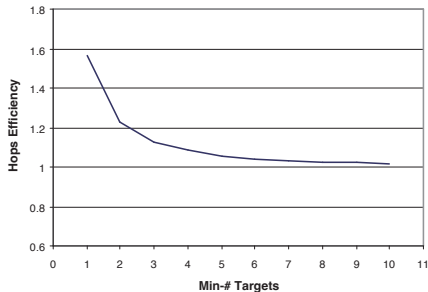




LTP – The impact of sampling several targets

Ideal Hops Efficiency for different number of targets

- # targets $\uparrow \Rightarrow C_h \rightarrow 1$
- 4 targets $\Rightarrow$ already very close to optimal



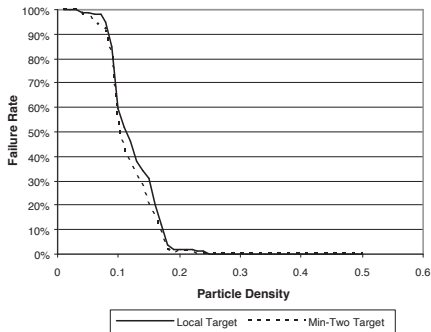


LTP – Failure rate

- A variety of sensor fields:
($n \in [100, 5000]$,
 $0.01 \leq d \leq 0.5$,
 $100m \times 100m$ square,
 $R = 5m$)
- for $d \leq 0.1$ both protocols
almost always backtrack
- for $d \geq 0.2$ the failure
rate drops very fast to 0.

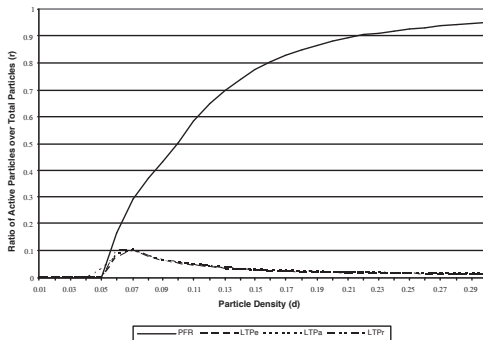
Failure rate for density $d \in [0.01, 0.5]$ and

$$\alpha = 90$$





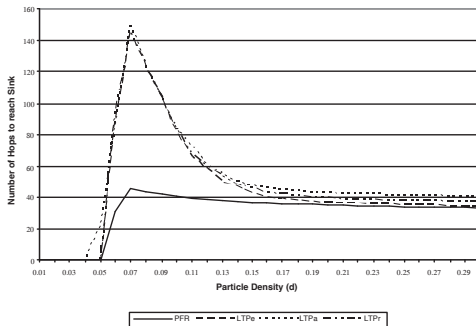
Ratio of activated particles over density



- PFR behaves very well ($r \leq 0.3$) for low densities ($d \leq 0.07$)
- PFR's energy dissipation increases with density
- LTP performs best in dense networks



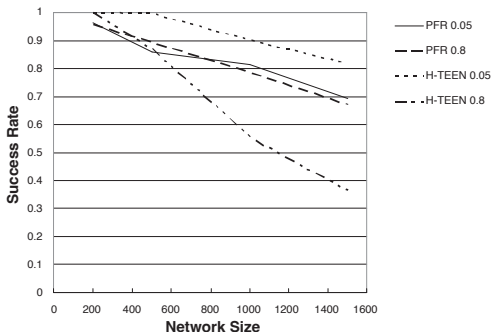
Backtracks over density



- For very low densities (i.e. $d \leq 0.12$), LTP backtracks a lot.
- As density increases, the number of backtracks of LTP reduces fast and almost reaches zero.



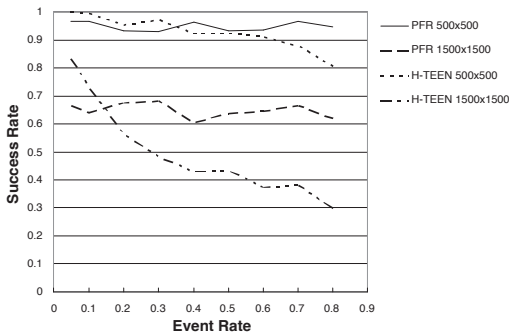
Success Rate over Network Size



- For small networks, $0.85 \leq \mathbb{P}_s \leq 1$.
- For large networks, success rate drops:
 - For SW-PFR $\mathbb{P}_s \rightarrow 0.7$.
 - For H-TEEN $\mathbb{P}_s \rightarrow 0.8$ for $l_s = 0.05$, and $\mathbb{P}_s \rightarrow 0.3$ for $l_s = 0.8$.



Success Rate over Injection Rate

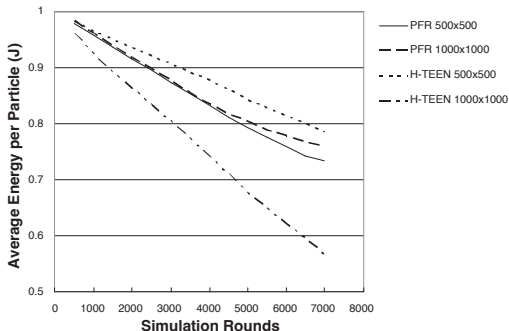


$$I_s \in [0.05, 0.8]$$

- For large networks the success rate of H-TEEN drops significantly when I_s increases.
- SW-PFR seems to be less affected by changes in I_s .



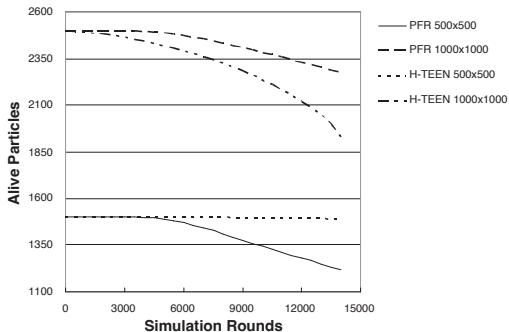
Average Energy over Time



- For small network size, H-TEEN is more energy efficient.
- For large network size, the energy consumption of H-TEEN increases dramatically.
- The energy consumption of SW-PFR is almost identical in both cases.



Number of Alive Particles over Time



- For small network size SW-PFR “drains” particles rapidly.
- For large network size H-TEEN “drains” particles rapidly.



Conclusions

- SW-PFR behaves quite well for all network sizes and injection rates.
- SW-PFR tends to *strain the particles which lie close to the sink* more.
- H-TEEN behaves very well on *small* network size and *low injection* rate settings, while it is “expensive” on large network areas and high injection rate.



The End

Questions?